

Finite Dimensional Vector Spaces (2005–2014)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS

19

TOTAL MARKS

88 M

TEST DURATION

57 Min

PAPER PATTERN

Classic Paper Pattern (2

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		19	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q13 • ID: JAM_2014_Q13)

MCQ

+2 M

Neg: -0.67

- Q.13 Consider the subspace $W = \{ (x_1, x_2, \dots, x_{10}) \in \mathbb{R}^{10} : x_n = x_{n-1} + x_{n-2} \text{ for } 3 \leq n \leq 10 \}$ of the vector space $\mathbb{R}^{10}$. The dimension of W is
- (A) 2 (B) 3 (C) 9 (D) 10

Question 2 (JAM 2014 • Q9 • ID: JAM_2014_Q9)

MCQ

+1 M

Neg: -0.33

- Q.9 Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by $T(x, y, z) = (x + y, y + z, z + x)$ for all $(x, y, z) \in \mathbb{R}^3$. Then
- (A) $\text{rank}(T) = 0$, $\text{nullity}(T) = 3$ (B) $\text{rank}(T) = 2$, $\text{nullity}(T) = 1$
(C) $\text{rank}(T) = 1$, $\text{nullity}(T) = 2$ (D) $\text{rank}(T) = 3$, $\text{nullity}(T) = 0$

Question 3 (JAM 2014 • Q8 • ID: JAM_2014_Q8)

MCQ

+1 M

Neg: -0.33

- Q.8 Which one of the following is a subspace of the vector space $\mathbb{R}^3$?
- (A) $\{ (x, y, z) \in \mathbb{R}^3 : x + 2y = 0, 2x + 3z = 0 \}$
(B) $\{ (x, y, z) \in \mathbb{R}^3 : 2x + 3y + 4z - 3 = 0, z = 0 \}$
(C) $\{ (x, y, z) \in \mathbb{R}^3 : x \geq 0, y \geq 0 \}$
(D) $\{ (x, y, z) \in \mathbb{R}^3 : x - 1 = 0, y = 0 \}$

Question 4 (JAM 2013 • Q10 • ID: JAM_2013_Q10)

MCQ

+2 M

Neg: -0.67

Q.10 Let $\mathcal{P}_n$ be the real vector space of all polynomials of degree at most n . Let $D: \mathcal{P}_n \rightarrow \mathcal{P}_{n-1}$ and $T: \mathcal{P}_n \rightarrow \mathcal{P}_{n+1}$ be the linear transformations defined by

$$D(a_0 + a_1x + a_2x^2 + \dots + a_nx^n) = a_1 + 2a_2x + \dots + na_nx^{n-1},$$

$$T(a_0 + a_1x + a_2x^2 + \dots + a_nx^n) = a_0x + a_1x^2 + a_2x^3 + \dots + a_nx^{n+1},$$

respectively. If A is the matrix representation of the transformation $DT - TD: \mathcal{P}_n \rightarrow \mathcal{P}_n$ with respect to the standard basis of $\mathcal{P}_n$, then the trace of A is

- (A) $-n$ (B) n (C) $n+1$ (D) $-(n+1)$

Question 5 (JAM 2013 • Q9 • ID: JAM_2013_Q9)

MCQ

+2 M

Neg: -0.67

Q.9 Let V be the vector space of all 2×2 matrices over $\mathbb{R}$. Consider the subspaces

$$W_1 = \left\{ \begin{pmatrix} a & -a \\ c & d \end{pmatrix} : a, c, d \in \mathbb{R} \right\} \text{ and } W_2 = \left\{ \begin{pmatrix} a & b \\ -a & d \end{pmatrix} : a, b, d \in \mathbb{R} \right\}.$$

If $m = \dim(W_1 \cap W_2)$ and $n = \dim(W_1 + W_2)$, then the pair (m, n) is

- (A) $(2, 3)$ (B) $(2, 4)$ (C) $(3, 4)$ (D) $(1, 3)$

Question 6 (JAM 2013 • Q1 • ID: JAM_2013_Q1)

MCQ

+2 M

Neg: -0.67

Q.1 Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \\ 1 & 5 & 3 \end{pmatrix}$ and V be the vector space of all $X \in \mathbb{R}^3$ such that $AX = 0$. Then $\dim(V)$

is

- (A) 0 (B) 1 (C) 2 (D) 3

Question 7 (JAM 2012 • Q11 • ID: JAM_2012_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Consider the following subspace of $\mathbb{R}^3$:

$$W = \left\{ (x, y, z) \in \mathbb{R}^3 \mid 2x + 2y + z = 0, 3x + 3y - 2z = 0, x + y - 3z = 0 \right\}.$$

The dimension of W is

- (A) 0 (B) 1 (C) 2 (D) 3

Question 8 (JAM 2012 • Q10 • ID: JAM_2012_Q10)

MCQ

+6 M

Neg: -2.0

Q.10 Let W be a vector space over $\mathbb{R}$ and let $T: \mathbb{R}^6 \rightarrow W$ be a linear transformation such that $S = \{Te_2, Te_4, Te_6\}$ spans W . Which one of the following must be TRUE?

- (A) S is a basis of W
- (B) $T(\mathbb{R}^6) \neq W$
- (C) $\{Te_1, Te_3, Te_5\}$ spans W
- (D) $\ker(T)$ contains more than one element

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Question 9 (JAM 2011 • Q13 • ID: JAM_2011_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 For $n \neq m$, let $T_1: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T_2: \mathbb{R}^m \rightarrow \mathbb{R}^n$ be linear transformations such that $T_1 T_2$ is bijective. Then

- (A) $\text{rank}(T_1) = n$ and $\text{rank}(T_2) = m$
- (B) $\text{rank}(T_1) = m$ and $\text{rank}(T_2) = n$
- (C) $\text{rank}(T_1) = n$ and $\text{rank}(T_2) = n$
- (D) $\text{rank}(T_1) = m$ and $\text{rank}(T_2) = m$

Question 10 (JAM 2011 • Q12 • ID: JAM_2011_Q12)

MCQ

+6 M

Neg: -2.0

Q.12 Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation, where $n \geq 2$. For $k \leq n$, let

$$E = \{v_1, v_2, \dots, v_k\} \subseteq \mathbb{R}^n \text{ and } F = \{Tv_1, Tv_2, \dots, Tv_k\}.$$

Then

- (A) If E is linearly independent, then F is linearly independent
- (B) If F is linearly independent, then E is linearly independent
- (C) If E is linearly independent, then F is linearly dependent
- (D) If F is linearly independent, then E is linearly dependent

Question 11 (JAM 2010 • Q15 • ID: JAM_2010_Q15)

MCQ

+6 M

Neg: -2.0

Q.15 Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear transformation whose matrix with respect to the standard basis of $\mathbf{R}^3$ is $\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$, where a, b, c are real numbers not all zero. Then T

- (A) is one-to-one
- (B) is onto
- (C) does not map any line through the origin onto itself
- (D) has rank 1

Question 12 (JAM 2010 • Q14 • ID: JAM_2010_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear transformation whose matrix with respect to the standard basis $\{e_1, e_2, e_3\}$ of $\mathbf{R}^3$ is $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. Then T

- (A) maps the subspace spanned by e_1 and e_2 into itself
- (B) has distinct eigenvalues
- (C) has eigenvectors that span $\mathbf{R}^3$
- (D) has a non-zero null space

Question 13 (JAM 2010 • Q11 • ID: JAM_2010_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Let $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be a linear transformation such that $T((1, 2)) = (2, 3)$ and $T((0, 1)) = (1, 4)$. Then $T((5, 6))$ is

- (A) $(6, -1)$
- (B) $(-6, 1)$
- (C) $(-1, 6)$
- (D) $(1, -6)$

Question 14 (JAM 2009 • Q4 • ID: JAM_2009_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 Let $T: \mathbf{R}^4 \rightarrow \mathbf{R}^4$ be a linear transformation satisfying $T^3 + 3T^2 = 4I$, where I is the identity transformation. Then the linear transformation $S = T^4 + 3T^3 - 4I$ is

- (A) one-one but not onto
- (B) onto but not one-one
- (C) invertible
- (D) non-invertible

Question 15 (JAM 2009 • Q1 • ID: JAM_2009_Q1)

MCQ

+6 M

Neg: -2.0

Q.1 Let V be the vector space of all 6×6 real matrices over the field $\mathbb{R}$. Then the dimension of the subspace of V consisting of all symmetric matrices is

- (A) 15 (B) 18 (C) 21 (D) 35

Question 16 (JAM 2008 • Q11 • ID: JAM_2008_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that $T(1, 2, 3) = (1, 2, 3)$, $T(1, 5, 0) = (2, 10, 0)$ and $T(-1, 2, -1) = (-3, 6, -3)$. The dimension of the vector space spanned by all the eigenvectors of T is

- (A) 0 (B) 1 (C) 2 (D) 3

Question 17 (JAM 2007 • Q5 • ID: JAM_2007_Q5)

MCQ

+6 M

Neg: -2.0

5. Which of the following sets is a basis for the subspace

$$W = \left\{ \begin{bmatrix} x & y \\ 0 & t \end{bmatrix} : x + 2y + t = 0, y + t = 0 \right\}$$

of the vector space of all real 2×2 matrices?

- (A) $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$
- (B) $\left\{ \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$
- (C) $\left\{ \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \right\}$
- (D) $\left\{ \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \right\}$

Question 18 (JAM 2006 • Q8 • ID: JAM_2006_Q8)

MCQ

+6 M

Neg: -2.0

8. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_1 - x_2, 0).$$

If $N(T)$ and $R(T)$ denote the null space and the range space of T respectively, then

- (A) $\dim N(T) = 2$
- (B) $\dim R(T) = 2$
- (C) $R(T) = N(T)$
- (D) $N(T) \subset R(T)$

Question 19 (JAM 2005 • Q7 • ID: JAM_2005_Q7)

MCQ

+6 M

Neg: -2.0

7. Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be a linear transformation and I be the identity transformation of $\mathbf{R}^3$. If there is a scalar c and a non-zero vector $x \in \mathbf{R}^3$ such that $T(x) = cx$, then $\text{rank}(T - cI)$

- (A) cannot be 0
- (B) cannot be 1
- (C) cannot be 2
- (D) cannot be 3

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Finite Dimensional Vector Spaces (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	A
2	MCQ	+1	D
3	MCQ	+1	A
4	MCQ	+2	C
5	MCQ	+2	B
6	MCQ	+2	B
7	MCQ	+6	B
8	MCQ	+6	D
9	MCQ	+6	D
10	MCQ	+6	B

Q. No.	Type	Marks	Official Key
11	MCQ	+6	C
12	MCQ	+6	C
13	MCQ	+6	A
14	MCQ	+6	D
15	MCQ	+6	C
16	MCQ	+6	D
17	MCQ	+6	D
18	MCQ	+6	A
19	MCQ	+6	D

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.